

NANDINI SAXENA

Curriculum Vitae

PERSONAL INFORMATION

Address: Pune, Maharashtra, India	Email: nandinisaxenawork@gmail.com
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Medium: medium.com/@nandinisaxenawork	Nationality: Indian

ABOUT ME

Final-year Computer Science bachelor's student (CGPA 9.3/10) focused on machine learning and data science, with hands-on research experience in navigation under sensor failure, synthetic data integrity, fraud detection under concept drift, and applied computer vision. Seeking a **Pflichtpraktikum** (approximately 16-18 weeks) in Berlin, **January to April 2027**, in any **data science, machine learning, or applied AI setting**, not tied to a single domain. Combines independent technical project work with team leadership experience running a 1500+ member developer community.

WORK EXPERIENCE

- Machine Learning Intern** — DIAT, Defence Research and Development Organisation (DRDO), Pune *Jan 2026 – Apr 2026*
- Built a transformer-based sequence model for UAV trajectory reconstruction during GPS outages using IMU sensor data, reducing trajectory RMSE by ~28% versus classical dead-reckoning integration
 - Designed the dead-reckoning pipeline end-to-end — noise filtering, bias correction, and drift modelling for inertial-only navigation under GPS denial
- Vice President** — Google Developer Groups (GDG) BVP Pune *Jun 2025 – Sept 2026*
- Lead a 12-person core team running technical programming for a 1500+ member developer community
 - Secured and manage 5+ industry partnerships for events, sponsorships, and speaker engagements
- Software Development Engineer Intern** — MCarbon Tech *Jun 2025 – Jul 2025*
- Built a smart ticketing system on Spring Boot, Flask, and Vue.js, covering backend services, API integration, and frontend delivery

EDUCATION AND TRAINING

- BTech, Computer Science & Engineering — CGPA 9.3/10** — Bharati Vidyapeeth (Deemed to be University), College of Engineering, Pune *Jul 2023 – June 2027*
- Relevant coursework: Machine Learning, Deep Learning, Natural Language Processing, Agentic AI, Business Intelligence, Cloud Computing
- One-year gap between higher secondary education and the start of this degree (2022–2023), spent preparing for undergraduate engineering entrance examinations.*
- Higher Secondary Certificate (Class XII) — 82.6%** — Delhi Public School, Hisar *April 2021 – March 2022*
- Secondary School Certificate (Class X) — 96%** — Delhi Public School, Hisar *April 2019 – March 2020*

PERSONAL SKILLS

Mother tongue: Hindi

Other languages

Language	Listening	Reading	Spoken interaction	Spoken production	Writing
English	C1	C1	C1	C1	C1
German	B1	B1	B1	B1	B1

Self-assessed against the Common European Framework of Reference for Languages (CEFR).

Communication skills: Strong written and verbal communication, developed through technical writing (published explainers on Medium) and leading partner/sponsor conversations as GDG Vice President.

Organisational / managerial skills: Lead a 12-person core team and coordinate programming for a 1500+ member community; manage 5+ concurrent industry partnerships end-to-end (outreach, negotiation, delivery).

Job-related skills

- Languages: Python, Java, SQL, JavaScript, C, TypeScript
- ML / DL: PyTorch, PyTorch Geometric, TensorFlow, HuggingFace Transformers, scikit-learn, XGBoost, Statsmodels, SHAP
- Focus areas: Time-series Forecasting, Graph Neural Networks, Sequence Modeling, Concept-drift Analysis, Computer Vision, Model explainability
- Backend / Infra: FastAPI, Flask, Spring Boot, PostgreSQL, Git, AWS, GCP

Digital skills

Information & data literacy	Communication & collaboration	Content creation	Safety	Problem solving
Proficient	Proficient	Proficient	Independent	Proficient

Self-assessed against the European Digital Competence Framework (DigComp).

ADDITIONAL INFORMATION

Selected projects (github.com/nandini1612)

- [GateIO](#) — Yaw-rate-gated causal TCN for UAV GPS-outage bridging: extended DRDO research, gating a causal temporal convolutional network with a yaw-rate signal to bridge GPS outages in UAV navigation
- Disentangling logical validity from statistical utility in synthetic tabular data: constraint-repair extension for hierarchical tabular generation models, enforcing logical/business constraints while preserving statistical utility
- [Phenological Mismatch Observatory](#): combined iNaturalist and eBird observation data to compute phenology mismatch scores, validated to 6.7-day / 2.5-day MAE against ground truth; now expanding into a CV classifier for citizen-science photos
- [Chicago Crime Analytics Platform](#): full-stack spatial-temporal analytics platform (Flask, React, PostGIS) processing 50,000+ real records, with SMA-based crime forecasting
- [AI Financial Crime Detector](#): compared GNN and XGBoost fraud detection under concept drift on the Elliptic Bitcoin dataset, quantifying performance degradation as transaction patterns shift over time
- Additional: [Fashion Colour vs. Economic Trend Analysis](#) (VAR forecasting) · [Conversational Analytics](#) (CSAT prediction, SHAP explainability) · [Neonatal EEG Seizure Detection](#) (in progress)

Publications (Medium, [@nandinisaxenawork](#))

- [Explainers on Prophet's seasonality decomposition](#), [GNNs vs. simpler models under distribution drift](#), [CNN inductive biases](#) and [Markov chains](#)

Leadership and activities

- Founder, OpportYouNity — youth opportunity & mental-health platform
- Head of PR, Null Student Chapter
- Classical Bharatanatyam dancer, 10+ years (RAQS College Dance Team)